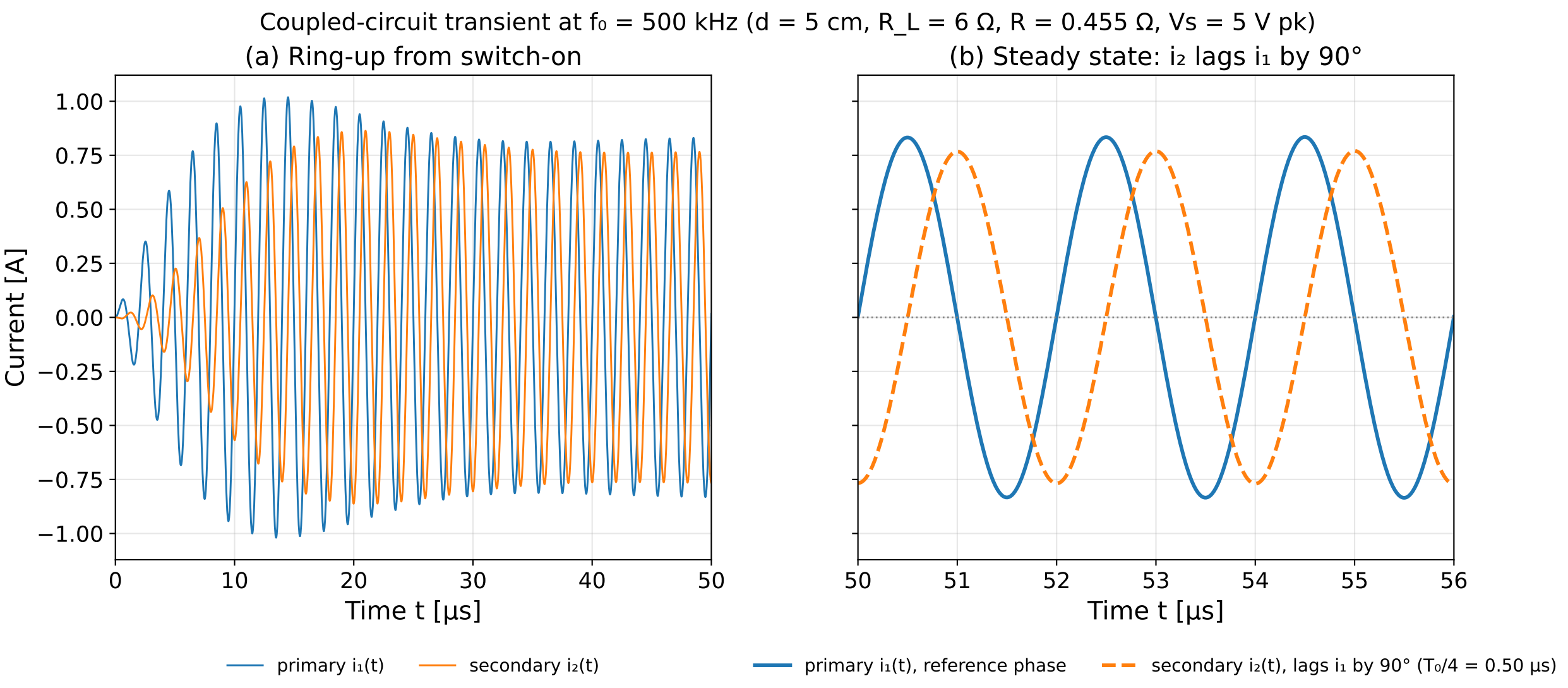
Coupled-circuit transient at f₀ = 500 kHz (d = 5 cm, R_L = 6 Ω, R = 0.455 Ω, Vs = 5 V pk)
(a) Ring-up from switch-on
(b) Steady state: i₂ lags i₁ by 90°
Current [A]
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
0
10
20
30
40
50
Time t [µs]
50
51
52
53
54
55
56
Time t [µs]
primary i₁(t)
secondary i₂(t)
primary i₁(t), reference phase
secondary i₂(t), lags i₁ by 90° (T₀/4 = 0.50 µs)